Reveal Optionality in Encrypted Mempools

Separation, Pricing, and Blind Simultaneous Release with Post-Quantum Time-Locks

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Abstract

Encrypted mempools hide transactions until their order is fixed, but every design must also say who opens the ciphertext afterwards, and whoever can decide after seeing new information holds a free option. We formalize where this option lives and what removing it costs. First, we split reveal optionality into an exogenous part driven by public information, which no mechanism can remove, and an endogenous part created by information the mechanism itself releases. We give an exact value and positivity criterion for a party with binary control, and show that sound and timely forced opening removes the endogenous part. Second, forced opening by public computation is feasible only if the adversary's speed advantage is below an explicit ratio of protocol times, which falls below two once a ciphertext waits one slot in a public mempool. Third, in commit-reveal designs that price non-reveal with a forfeitable fee, an attacker with two preceding commitments makes wrong guesses nearly free, so deterrence requires a fee of at least half the attacker's gain when its best guess has probability above $2/(F+1)$. Fourth, threshold decryption and key-publisher allowlists move the option to every coalition able to block decryption, and no threshold keeps both early decryption and blocking beyond about half the committee. Fifth, we give a construction, blind simultaneous release, that escrows every release decision under a short post-quantum time-lock so that it is taken before any plaintext is visible. Assuming honest solvers are within a factor of 5 (or 11) of the adversary's sequential speed under current Ethereum timing, and with a deposit forfeited only on verifiable inconsistency, it removes the endogenous option and, if misdirected executions lose at least the fees they save, restores the single-commitment fee. We calibrate to Ethereum parameters and machine-check the core statements in Lean 4.

1 Introduction

Transactions on public blockchains wait in a broadcast queue, the mempool, before a block producer orders them. Anyone watching the queue can insert a transaction ahead of a large trade and profit from its price impact. Encrypted mempools address this by hiding transaction contents until the ordering is fixed [39, 26, 21, 41]. Ethereum is actively evaluating several designs: a commit-reveal design in which senders or their key publishers release decryption keys after ordering (LUCID, EIP-8184 [32]), a scheme-agnostic framework with registered key providers (EIP-8105 [24]), threshold decryption by validator committees [29, 30], and hybrids of these. Threshold-decrypted mempools have also been deployed outside Ethereum [4].

All of these designs face the same structural question: once the ciphertext is committed and ordered, who can cause it to be opened, and when do they decide? If a party can decide after the ordering is fixed and after new information has arrived, it holds an option. A sender who sees prices move against its trade can decline to reveal; a key publisher who sees other keys can withhold its own; a committee subset can refuse to release its shares. The option is free unless the protocol

charges for it. This problem has been identified as the main weakness of commit-reveal designs [28] and as the subject of open design questions in the ongoing Ethereum process, which notes that deterrence by penalties is hard because it is unclear how to quantify the value of the option [46]. An analogous option for block builders under enshrined proposer-builder separation has been measured empirically [27].

This paper gives a formal account of the option: how to measure it, which mechanisms remove it, and where it goes when it is removed from the sender.

Contributions.

1. **Measure and separation (Section 4).** Reveal optionality relative to an information filtration, split into an exogenous and an endogenous part, with an exact value and positivity criterion for binary control and a bound under sound timely forced opening (Theorem 1).

2. **Timing feasibility (Section 5).** Forced opening by public computation is feasible if and only if the adversary's speed advantage is at most an explicit ratio of protocol times, which is below two after one slot in a public mempool (Theorem 2).

3. **Fee inversion (Section 6).** With two preceding commitments an attacker makes wrong guesses nearly free in LUCID-style pricing, so the deterring fee is at least $\Lambda/2$ once its best guess has probability above $2/(F+1)$ (Theorem 3).

4. **Option migration (Section 7).** Threshold release and allowlists give the option to blocking coalitions, no threshold keeps both harms beyond about half the committee, and the penalty must exceed the largest single gain (Theorem 4); hybrid release leaks the option back through correlated outages (Corollary 3).

5. **A construction (Section 8).** Blind simultaneous release, a post-quantum adaptation of commit-reveal-recover techniques to key and share release (Theorem 5).

6. **Calibration and machine-checked proofs (Section 9, Appendix B).** Option values and fees at current ETH volatility, and Lean 4 proofs of the finite-probability and arithmetic core.

Consequences for current designs. Commit-reveal pricing of non-reveal is defeated by fee inversion unless the victim limits preceding commitments to one or posts a fee close to the extractable value. For allowlisted publishers and threshold committees, withholding is not merely an availability attack: a withholder that exercises rarely, in high-value states, looks like downtime, so the penalty must be sized to the largest single gain. Ticket-first threshold designs [29] satisfy our timing criterion for the sender, and their residual option sits with blocking coalitions.

2 Background

Encrypted mempools. A sender encrypts a transaction, the ciphertext is included and ordered, and the plaintext is released for execution afterwards. Designs differ in who can release it.

- *Sender or publisher release.* In LUCID [32], a sealed transaction is committed by a ticket and ordered by a top-of-block (ToB) fee; a key publisher, possibly the sender itself, releases the key after the ordering is fixed. A revealed transaction is refunded $(F-1)/F$ of its ToB fee, with $F = 128$; a non-revealed one forfeits the fee. Each transaction sets a cap

`max_preceding_commitments` on the number of commitments that may execute before it; the specification counts each committed transaction of a multi-key bundle as a commitment whether or not it reveals, and we count every included commitment ahead of the victim in the same way. EIP-8105 [24] lets any account register a key provider, allows providers to withhold keys at their discretion, and records which keys were released; recent proposals restrict providers to a validator-approved allowlist [46, 25].

- *Timed release.* A time-lock puzzle or verifiable delay function lets anyone recover the plaintext after a computational delay [38, 9, 44, 42].
- *Threshold release.* A committee holds shares of a decryption key; any t of n shares suffice [21, 40, 5, 43, 29].
- *Hybrid release.* The committee normally decrypts, and the sender can self-decrypt if the committee is unavailable [30].

The free option. A party who can decide after the ordering is fixed whether a committed transaction executes holds an option on any information that arrives in between. The analogous option of block builders under enshrined proposer-builder separation has been analysed by Mazorra et al. [27], who show that its value grows with volatility and window length and that, with an 8-second window, it would have been profitable to exercise in 0.82% of historical blocks overall and in up to 6% on high-volatility days. For encrypted mempools the option has been described informally [28, 2, 41]; this paper makes it precise.

3 Model

Timeline. A party commits at time t_0 (the ciphertext or a hiding commitment becomes public and binding). The ordering and the set of included items are fixed at $t_1 > t_0$. Execution is determined at $t_2 > t_1$. We write $\delta = t_1 - t_0$ and $\Delta = t_2 - t_1$.

Information. On a probability space $(\Omega, \Sigma, \Pr)$ there are two filtrations: $\mathcal{G} = (\mathcal{G}_t)$, exogenous public information (prices on external venues, oracle updates), and $\mathcal{H} = (\mathcal{H}_t)$, information released by the mechanism (other parties' plaintexts, keys or key shares, reveal and withhold decisions). Let $\mathcal{F}_t = \mathcal{G}_t \vee \mathcal{H}_t$. We assume *privacy before ordering*: $\mathcal{H}_t$ is trivial for $t < t_1$, so $\mathcal{F}_{t_0} = \mathcal{G}_{t_0}$. This isolates optionality from frontrunning, which is a failure of privacy and is studied separately.

Payoffs. For the party under study, U_1 is its payoff if the committed item executes at its committed position and U_0 its payoff otherwise, both net of protocol charges and integrable. Put $D = U_0 - U_1$, the gain from non-execution, and $D_t^{\mathcal{K}} = \mathbb{E}[D \mid \mathcal{K}_t]$. For an execution indicator $X \in \{0, 1\}$, $U_X = XU_1 + (1 - X)U_0 = U_1 + (1 - X)D$.

Mechanisms. Fix the strategies of all other parties. A mechanism M maps each admissible post-commitment strategy σ of the party to an execution indicator X^σ. For a filtration $\mathcal{K}$, $\mathcal{A}_M(\mathcal{K})$ denotes the indicators obtainable by strategies using only $\mathcal{K}$-information, and $\mathcal{A}_M(t_0)$ those obtainable by strategies fixed at t_0. The passive strategy, which takes no action after t_0, is always admissible and belongs to $\mathcal{A}_M(t_0)$.

Definition 1 (Reveal optionality)**.** The optionality of the party under M relative to $\mathcal{K}$ is

$$V^{\mathcal{K}}(M) = \sup_{X \in \mathcal{A}_M(\mathcal{K})} \mathbb{E}[U_X] - \sup_{X \in \mathcal{A}_M(t_0)} \mathbb{E}[U_X].$$

The total, exogenous and endogenous optionality are $V^{\mathrm{tot}} = V^{\mathcal{F}}$, $V^{\mathrm{exo}} = V^{\mathcal{G}}$ and $V^{\mathrm{endo}} = V^{\mathcal{F}} - V^{\mathcal{G}}$. All three are nonnegative because $\mathcal{A}_M(t_0) \subseteq \mathcal{A}_M(\mathcal{G}) \subseteq \mathcal{A}_M(\mathcal{F})$.

The exogenous option cannot be removed by mechanism design alone: a transaction can make its own effect depend on public state at execution time, which gives its author the same exposure to public information. The endogenous option is created by the mechanism, by releasing information to a party before that party's last decision. Our design criteria target the endogenous option and treat the exogenous one as a quantity to be priced.

Definition 2 (Sound timely forced opening)**.** M has $(\varepsilon_{\mathrm{live}}, \varepsilon_{\mathrm{snd}})$-sound timely forced opening (STFO) for a committing party if there is a deterministic function Open of the public transcript c at t_0, valued in messages $\cup\{\bot\}$, and for every admissible strategy σ of the party an event Γ_σ with $\Pr[\Gamma_\sigma^c] \le \varepsilon_{\mathrm{live}} + \varepsilon_{\mathrm{snd}}$, such that:

1. *Timeliness:* except with probability $\varepsilon_{\mathrm{live}}$, other parties compute $\mathsf{Open}(c)$ by t_2 without input from the committing party after t_0;
2. *Soundness:* except with probability $\varepsilon_{\mathrm{snd}}$, no opening other than $\mathsf{Open}(c)$ is accepted, whatever the committing party does;
3. *Execution rule:* on Γ_σ, $X^\sigma = \bar{X}$, where $\bar{X}$ depends only on $\mathsf{Open}(c)$ and other parties' actions, and not on σ.

Following Ambrona, Beunardeau and Toledo [42], the definition allows $\mathsf{Open}(c) = \bot$: committing to a malformed message is a decision fixed at t_0 and carries no option. What matters is that the outcome is fixed by the commitment and computable without the committer. Their observation that a committer able to forge opening proofs can later choose between m and $\bot$ is precisely a failure of soundness in this sense.

Definition 3 (Binary control)**.** A party has *binary control on R at t^** if $R \in \mathcal{G}_{t^*}$ for some $t^* \in [t_0, t_2]$ and the obtainable indicators are exactly those of the form $X = X_0 \mathbf{1}_{R^c} + (1-Y)\mathbf{1}_R$, where X_0 is not influenced by the party and $Y \in \{0,1\}$ is $\mathcal{K}_{t^*}$-measurable under a $\mathcal{K}$-strategy and $\mathcal{F}_{t_0}$-measurable under a committed strategy.

4 Separation and the value of the option

Lemma 1 (Optimal binary choice)**.** *Let Z be integrable and $\mathcal{C}$ a sub-σ-algebra. Then* $\sup\{\mathbb{E}[YZ] : Y \in \{0,1\}$ *$\mathcal{C}$-measurable$\} = \mathbb{E}[(\mathbb{E}[Z \mid \mathcal{C}])^+]$, attained by $Y = \mathbf{1}\{\mathbb{E}[Z \mid \mathcal{C}] > 0\}$.*

Proof. For $\mathcal{C}$-measurable $Y \in \{0,1\}$, $\mathbb{E}[YZ] = \mathbb{E}[Y\mathbb{E}[Z \mid \mathcal{C}]] \le \mathbb{E}[(\mathbb{E}[Z \mid \mathcal{C}])^+]$, with equality for the stated Y. □

Lemma 2 (Strict conditional Jensen for x^+)**.** *Let $\mathcal{C} \subseteq \mathcal{C}'$ and let W be integrable and $\mathcal{C}'$-measurable, with $m = \mathbb{E}[W \mid \mathcal{C}]$. Then $\mathbb{E}[W^+] \ge \mathbb{E}[m^+]$, with strict inequality if and only if*

$$\Pr\big[W > 0 \ge m\big] + \Pr\big[W < 0 \le m\big] > 0.$$

Proof. $W^+ \geq W$ and $W^+ \geq 0$ give $\mathbb{E}[W^+ \mid \mathcal{C}] \geq m^+$. The gap $\mathbb{E}[W^+ \mid \mathcal{C}] - m^+$ equals $\mathbb{E}[W^- \mid \mathcal{C}]$ on $\{m \geq 0\}$ and $\mathbb{E}[W^+ \mid \mathcal{C}]$ on $\{m < 0\}$. On $\{m \geq 0\}$ it is positive iff $\Pr[W < 0 \mid \mathcal{C}] > 0$, i.e. iff $\{W < 0 \leq m\}$ has positive conditional probability; on $\{m < 0\}$ it is positive iff $\Pr[W > 0 \mid \mathcal{C}] > 0$, i.e. iff $\{W > 0 \geq m\}$ has positive conditional probability (on $\{m = 0\}$ either event works, since a zero conditional mean with positive conditional mass on one side forces positive conditional mass on the other). Integrating gives the claim. □

The criterion uses non-strict inequalities on the side of m. With strict inequalities it would fail when $m = 0$: if $D = \pm 1$ with equal probability, $\mathcal{G}$ is trivial and the mechanism reveals D, the endogenous option equals $\mathbb{E}[D^+] = 1/2$ while $\{D > 0 > 0\}$ and $\{D < 0 < 0\}$ are empty.

Theorem 1 (Separation)**.** *(a) If M has $(\varepsilon_{\text{live}}, \varepsilon_{\text{snd}})$-STFO, $\varepsilon = \varepsilon_{\text{live}} + \varepsilon_{\text{snd}}$, and $D \in L^p$ for some $p \in [1, \infty]$, then for every filtration $\mathcal{K}$,*

$$V^{\mathcal{K}}(M) \leq \|D\|_p \, (2\varepsilon)^{1-1/p},$$

and the factor 2 may be dropped when Γ_σ does not depend on σ. In particular exact STFO gives $V^{\text{tot}} = V^{\text{exo}} = V^{\text{endo}} = 0$.

(b) If the party has binary control on R at t^* and $\mathcal{K} \in \{\mathcal{G}, \mathcal{F}\}$, then

$$V^{\mathcal{K}} = \mathbb{E}\big[(\mathbf{1}_R D^{\mathcal{K}}_{t^*})^+\big] - \mathbb{E}\big[(\mathbb{E}[\mathbf{1}_R D \mid \mathcal{F}_{t_0}])^+\big], \qquad V^{\text{endo}} = \mathbb{E}\big[\mathbf{1}_R (D^{\mathcal{F}}_{t^*})^+\big] - \mathbb{E}\big[\mathbf{1}_R (D^{\mathcal{G}}_{t^*})^+\big] \geq 0,$$

and $V^{\text{endo}} > 0$ if and only if $\Pr[R,\ D^{\mathcal{F}}_{t^*} > 0 \geq D^{\mathcal{G}}_{t^*}] + \Pr[R,\ D^{\mathcal{F}}_{t^*} < 0 \leq D^{\mathcal{G}}_{t^*}] > 0$.

Proof. (a) Let σ be a $\mathcal{K}$-strategy and σ_{pass} the passive strategy (no action after t_0), which is committed. On $\Gamma = \Gamma_\sigma \cap \Gamma_{\sigma_{\text{pass}}}$ both indicators equal $\bar{X}$, so

$$\mathbb{E}[U_{X^\sigma}] - \mathbb{E}[U_{X^{\text{pass}}}] = \mathbb{E}\big[(X^{\text{pass}} - X^\sigma) D \mathbf{1}_{\Gamma^c}\big] \leq \mathbb{E}[|D| \mathbf{1}_{\Gamma^c}] \leq \|D\|_p \Pr[\Gamma^c]^{1-1/p}$$

by Hölder's inequality, and $\Pr[\Gamma^c] \leq 2\varepsilon$ by the union bound ($\leq \varepsilon$ if the two events coincide). Since the committed supremum is at least $\mathbb{E}[U_{X^{\text{pass}}}]$, taking the supremum over σ proves the bound.

(b) Write $\mathbb{E}[U_X] = \mathbb{E}[U_{X_0} \mathbf{1}_{R^c}] + \mathbb{E}[\mathbf{1}_R U_1] + \mathbb{E}[\mathbf{1}_R (1 - X) D]$; only the last term depends on the party. Under a $\mathcal{K}$-strategy, $Y = 1 - X$ is a $\mathcal{K}_{t^*}$-measurable indicator on R, and $R \in \mathcal{G}_{t^*} \subseteq \mathcal{K}_{t^*}$, so Lemma 1 with $Z = \mathbf{1}_R D$ and $\mathcal{C} = \mathcal{K}_{t^*}$ gives the maximum $\mathbb{E}[(\mathbf{1}_R D^{\mathcal{K}}_{t^*})^+]$. Under a committed strategy the same lemma with $\mathcal{C} = \mathcal{F}_{t_0}$ gives $\mathbb{E}[(\mathbb{E}[\mathbf{1}_R D \mid \mathcal{F}_{t_0}])^+]$. Subtracting gives $V^{\mathcal{K}}$, and the committed terms cancel in V^{endo}. Since $R \in \mathcal{G}_{t^*}$, $(\mathbf{1}_R D^{\mathcal{K}})^+ = \mathbf{1}_R (D^{\mathcal{K}})^+$. For the sign apply Lemma 2 with $W = \mathbf{1}_R D^{\mathcal{F}}_{t^*}$ and $\mathcal{C} = \mathcal{G}_{t^*}$, using $\mathbb{E}[W \mid \mathcal{G}_{t^*}] = \mathbf{1}_R D^{\mathcal{G}}_{t^*}$. □

Corollary 1 (Timing criterion)**.** *Under binary control, if $\mathcal{H}_{t^*}$ is trivial, that is, the party's last decision precedes any release of mechanism information, then $V^{\text{endo}} = 0$.*

Corollary 2 (Hybrid release with independent outages)**.** *Suppose the committee is unavailable on an event R with $\Pr[R] = 1 - q$, the sender self-decrypts only on R, $\mathcal{G}_{t^*} = \mathcal{G}'_{t^*} \vee \sigma(R)$ with $\mathcal{F}_{t_0} \subseteq \mathcal{G}'_{t^*}$, and R is independent of $\sigma(D) \vee \mathcal{G}'_{t^*} \vee \mathcal{H}_{t^*}$. Let $V^{\text{endo}}_{\text{sr}}$ and $V^{\text{tot}}_{\text{sr}}$ denote the endogenous and total optionality of the same sender under pure sender release (binary control on Ω) with information $\mathcal{G}'$ and $\mathcal{H}$. Then $V^{\text{endo}} = (1 - q) V^{\text{endo}}_{\text{sr}}$ and $V^{\text{tot}} = (1 - q) V^{\text{tot}}_{\text{sr}}$.*

Proof. By independence, conditioning on $\sigma(R)$ does not change the conditional expectations of D, so $\mathbf{1}_R D^{\mathcal{K}}_{t^*} = \mathbf{1}_R D'^{\mathcal{K}}_{t^*}$ with D' independent of R, and $\mathbb{E}[\mathbf{1}_R (D'^{\mathcal{K}})^+] = (1 - q)\mathbb{E}[(D'^{\mathcal{K}})^+]$. Likewise $\mathbb{E}[\mathbf{1}_R D \mid \mathcal{F}_{t_0}] = (1 - q)\mathbb{E}[D \mid \mathcal{F}_{t_0}]$. Apply Theorem 1(b) to both mechanisms. □

Adjoining $\sigma(R)$ to the exogenous filtration treats the outage as part of the baseline: both terms of V^{endo} condition on it, so what is measured is the value of the mechanism's information given that the sender is pivotal. This is appropriate when outages are driven by infrastructure conditions outside the party's control. If a pivotal event were generated by a strategic act within the mechanism, such as a coalition withholding in response to plaintexts, then $R \notin \mathcal{G}_{t^*}$ and Theorem 1(b) does not apply; Section 7 treats that case by giving the coalition binary control on the whole space.

Lemma 3 (Gaussian closed form)**.** *If, conditionally on $\mathcal{C}$, $D^{\mathcal{F}}_{t^*} \sim \mathcal{N}(\mu, s^2)$ with $s > 0$, then $\mathbb{E}[(D^{\mathcal{F}}_{t^*})^+ \mid \mathcal{C}] = s\varphi(\mu/s) + \mu\Phi(\mu/s)$, and the conditional option value is $s\varphi(\mu/s) + \mu\Phi(\mu/s) - \mu^+$, the Bachelier call price net of intrinsic value.*

Proof. This is the classical Bachelier formula [1]; we include the short derivation. Write $D = \mu + sZ$ with Z standard normal. Then $\mathbb{E}[D^+] = \int_{-\mu/s}^{\infty} (\mu + sz)\varphi(z)\,dz = \mu\,(1 - \Phi(-\mu/s)) + s\varphi(\mu/s) = \mu\Phi(\mu/s) + s\varphi(\mu/s)$, using $\varphi' = -z\varphi$ and the symmetry of φ and Φ. The conditional option value is this quantity minus the committed value μ^+. □

Applications.

- *Sender or self-publisher release* (LUCID with the sender as key publisher; an EIP-8105 provider registered by the sender, which the permissionless registry allows). The sender has binary control on Ω with its last decision at the key deadline, after other keys are visible, so $\mathcal{H}_{t^*}$ is nontrivial and the endogenous option is positive whenever earlier releases can change the sign of the sender's expected gain.
- *Ticket-first threshold release* [29]. The sender commits to a ticket, publishes the ciphertext only after the ticket is included, and keys are released only after the set of available ciphertexts is fixed in consensus. The sender's last decision (publish or not) precedes any key release, so by Corollary 1 its endogenous option is zero; its exogenous option remains.
- *Timed commitments with forgeable proofs* [42]. Soundness fails and the committer has binary control at the opening time.
- *Hybrid release* [30]. Corollary 2 under independence; Section 7 treats the general case.

5 Forced opening by public computation

The inputs to a forced opening are of three kinds: data public when the ciphertext is first broadcast, secret state held by other parties (key shares, an enclave key), and data that becomes available only later (a beacon output, a certificate used as a witness). The second kind is the committee or hardware setting of Section 7. The third requires some party or consensus event to produce the later data, which in proof-of-stake reduces to trust in that party or in a supermajority [2]. This section treats the first kind.

Definition 4 (Public-computation forced opening)**.** $\mathsf{Open}(c)$ is computed by a public algorithm whose inputs are all public at the broadcast time $t_{\text{pub}} \le t_0$. Its running time for the fastest honest party relied on for timeliness is the difficulty $x > 0$; that party starts at $t_h \ge t_{\text{pub}}$. The adversary runs the same computation in time x/γ and may start at t_{pub}.

The adversary can always run the honest procedure. Privacy requires that it not finish before the ordering is fixed, and timeliness requires that honest parties finish by the execution deadline:

$$\mathsf{Private}(x):\ t_{\text{pub}} + x/\gamma \ge t_1, \qquad \mathsf{Timely}(x):\ t_h + x \le t_2.$$

δ (s)	Δ (s)	$w = 0$	$w = 12$	$w = 60$
1	8	9.0 / 9.0	0.69 / 1.62	0.15 / 1.13
1	12	13.0 / 13.0	1.00 / 1.92	0.21 / 1.20
2	8	5.0 / 5.0	0.71 / 1.57	0.16 / 1.13
2	12	7.0 / 7.0	1.00 / 1.86	0.23 / 1.19
4	8	3.0 / 3.0	0.75 / 1.50	0.19 / 1.13
4	12	4.0 / 4.0	1.00 / 1.75	0.25 / 1.19

Table 1: Feasibility ceiling $\gamma_{\max}$ with honest solvers starting at inclusion / at broadcast, for illustrative timings.

Theorem 2 (Timing feasibility)**.** *Assume $t_{\text{pub}} < t_1$ and $\gamma > 0$.*

(a) Some difficulty $x > 0$ satisfies $\mathsf{Private}(x)$ and $\mathsf{Timely}(x)$ if and only if

$$\gamma \le \gamma_{\max} := \frac{t_2 - t_h}{t_1 - t_{\text{pub}}}.$$

(b) Let $w = t_0 - t_{\text{pub}} \ge 0$ be the time the ciphertext waits before inclusion, and $\delta, \Delta > 0$. Then $\gamma_{\max} = (\delta + \Delta)/(\delta + w)$ if honest parties start at inclusion ($t_h = t_0$), and $\gamma_{\max} = (\delta + \Delta + w)/(\delta + w) = 1 + \Delta/(\delta + w)$ if they start at broadcast ($t_h = t_{\text{pub}}$). Both are strictly decreasing in w; the second tends to 1, and for $\gamma > 1$ it is below γ exactly when $w > (\delta + \Delta - \gamma\delta)/(\gamma - 1)$. In particular $\gamma_{\max} < 2$ whenever $\Delta < \delta + w$.

(c) If γ has distribution function G and the honest running time is ηx with η of distribution function H, then the probability of early opening $1 - G(x/(t_1 - t_{\text{pub}}))$ is non-increasing in x, the probability of late opening $p_{\text{late}}(x) = 1 - H((t_2 - t_h)/x)$ is non-decreasing in x, and with sound opening proofs the total option is at most $\|D\|_\infty\, p_{\text{late}}(x)$ by Theorem 1(a), provided $t_h \le t_2$.

Proof. (a) $\mathsf{Private}(x) \iff x \ge \gamma(t_1 - t_{\text{pub}})$ and $\mathsf{Timely}(x) \iff x \le t_2 - t_h$; take $x = \gamma(t_1 - t_{\text{pub}})$ for sufficiency.

(b) Substitute $t_{\text{pub}} = t_0 - w$, $t_1 = t_0 + \delta$, $t_2 = t_0 + \delta + \Delta$. The threshold is the solution of $(\gamma - 1)(\delta + w) > \Delta$, and $1 + \Delta/(\delta + w) < 2$ iff $\Delta < \delta + w$.

(c) G and H are non-decreasing; in (c) the good event is that honest solvers finish in time, which does not depend on the committing party's strategy. □

Discussion. The inequality is elementary, but it applies to every puzzle and delay function because the adversary may run the honest procedure. If a ciphertext waits one slot ($w = 12$ s) in a public mempool, the ceiling of Table 1 is below 2 in every configuration, while reported sequential speedups are larger: 20× for MinRoot [34], and a 10× advantage is tolerable only for some applications [44]. Timed release is therefore incompatible with ciphertexts that wait in a public mempool. If only a hiding commitment waits and the ciphertext is published after an inclusion slot is reserved, $w \approx 0$ and $\gamma_{\max} = 1 + \Delta/\delta$, between 3 and 13. Funding fast solvers to reduce γ makes timeliness depend on them, which is the setting of Section 7. Two qualifications apply to Theorem 2(c): $1 - G(\cdot)$ is the probability of a privacy breach, separate from the post-ordering option bounded by p_{late}; and the broadcast-start column is an optimistic ceiling when $w > 0$, since solvers would compute on unincluded traffic.

6 Fee inversion in commit-reveal pricing

We now study the main pricing instrument of commit-reveal designs. In LUCID [32], each sealed transaction carries a ToB fee f; commitments execute in order of decreasing ToB fee; a revealed transaction pays f/F and a non-revealed one forfeits f, with $F = 128$; and each transaction sets a cap m on the commitments that may execute before it. The fee prices *probabilistic frontrunning*: an attacker commits to a transaction that front-runs its guess of the victim's plaintext and reveals only if the guess was right [28]. The specification notes that with m preceding commitments an attacker can select among 2^{m-1} patterns by using $m-1$ commitments as selectors and the last to execute the chosen pattern, and it has been argued that an attacker who does not reveal must forfeit a large fee [28]. We show that the attacker can arrange to forfeit the large fee only when its guess is right.

Model. A victim's sealed transaction has ToB fee f and cap m. The attacker posts $k \le m$ commitments, each with fee f, the infimum that precedes the victim. The victim's plaintext lies in one of finitely many *guess classes*, sets of plaintexts against which a single prepared pattern extracts value, with probabilities $p_1 \ge p_2 \ge \ldots$, $\sum_t p_t \le 1$. We assume:

A1 (late reveal) the attacker learns the victim's class before its own reveal decision;

A2 (gains) executing the pattern prepared for the true class gains at most $\Lambda > 0$; executing a pattern prepared for another class is never more profitable for the attacker than aborting, that is, a misdirected execution loses at least the fees it would save (this is the premise of any non-reveal penalty: without it an attacker would reveal on every miss);

A3 (programmable commitments) a commitment's program can read state written by earlier transactions in the same block, and nothing about the plaintexts of later ones.

An *attack plan* (k, A, σ, a) consists of a set A of targeted classes, an injective map σ from A to nonempty reveal sets $S \subseteq [k]$ (the empty set executes nothing and cannot attack), and an abort reveal set $a \notin \sigma(A)$ used on a miss. Revealing exactly S costs $c(S) = f\big((k - |S|) + |S|/F\big)$. The expected profit is

$$\Pi = \sum_{t \in A} p_t\big(\Lambda - c(\sigma(t))\big) - \Big(1 - \sum_{t \in A} p_t\Big)\, c(a).$$

This *general* class lets every nonempty reveal set carry an arbitrary program, which over-approximates what sequentially executed commitments can do; its profit bounds that of every attacker from above. The *selector* class consists of plans implementable under A3 with designated selector and executor commitments: $k-1$ selectors and one executor e, where every targeted reveal set contains e (so there are 2^{k-1} attack sets) and the abort set is any other reveal set, either without e or with e alongside a selector pattern that makes it do nothing. Let f^*_m denote the least fee at which no plan with $k \le m$ is profitable.

Theorem 3 (Fee inversion)**.** *Assume $F > 1$ and $0 < p_1 < 1$. Let*

$$f_1(p) = \frac{p\Lambda}{(1-p) + p/F}, \qquad f_{\mathrm{inv}}(p) = \frac{p\Lambda}{p(1 + 1/F) + 2(1-p)/F}.$$

(a) With one commitment, every plan is unprofitable if and only if $f \ge f_1(p_1)$; so $f^*_1 = f_1(p_1)$ in both classes.

p_1	$f_1(p_1)$	$f_{\text{inv}}(p_1)$	$f_1(p_1)/p_1$	f_{inv}/f_1
1/256	0.0039	0.200	1.004	51
1/64	0.0159	0.502	1.016	32
1/16	0.0666	0.805	1.066	12
1/4	0.332	0.948	1.330	2.9

Table 2: Deterring ToB fee in units of the attacker's gain Λ, $F = 128$: the single-commitment threshold, the lower and upper bounds for caps $m \geq 2$, and the ratio of the lower bound to the single-commitment threshold.

(b) For every $m \geq 2$, the following selector plan is profitable whenever $f < f_{\text{inv}}(p_1)$. Commit a selector s and an executor e. On a hit, reveal only e, which executes the attack because s did not run. On a miss, reveal both; the executor observes that s ran and does nothing. Hence $f_m^* \geq \max\{f_1(p_1), f_{\text{inv}}(p_1)\}$.

(c) For every m and every plan in the general class, $\Pi \leq 0$ whenever $f \geq \Lambda/\big((1-p_1)+p_1/F\big) = f_1(p_1)/p_1$. Hence $f_m^* \leq f_1(p_1)/p_1$.

(d) $f_{\text{inv}}(p_1) \geq \Lambda/2$ if and only if $p_1 \geq 2/(F+1)$.

Proof. (a) With $k = 1$ the reveal sets are $\{1\}$, of cost f/F, and $\emptyset$, of cost f, which can only abort. By injectivity at most one class is targeted, through $\{1\}$, with abort $\emptyset$, giving $\Pi = p_t(\Lambda - f/F) - (1-p_t)f$; the plan with no target has $\Pi = -c(a) \leq 0$. Since f_1 is increasing in p, every plan is unprofitable iff $p_1(\Lambda - f/F) - (1-p_1)f \leq 0$, i.e. iff $f \geq f_1(p_1)$.

(b) A hit costs $c(\{e\}) = f(1+1/F)$ (the selector is forfeited, the executor refunded) and a miss costs $c(\{s,e\}) = 2f/F$. Targeting the top class gives $\Pi = p_1(\Lambda - f(1+1/F)) - (1-p_1)\,2f/F$, which is positive iff $f < f_{\text{inv}}(p_1)$. The plan uses two commitments and so is available under every cap $m \geq 2$.

(c) Write $\hat{c}(S) = c(S)/f = (k - |S|) + |S|/F$, so $\hat{c}(S) \geq 1$ whenever $|S| < k$ and $\hat{c}([k]) = k/F$. Then $\Pi = \Lambda P_A - f\kappa$ with $P_A = \sum_{t\in A} p_t \leq 1$ and $\kappa = \sum_{t\in A} p_t \hat{c}(\sigma(t)) + (1-P_A)\hat{c}(a)$. If $a = [k]$, every attack set has $\hat{c} \geq 1$, so $\kappa \geq P_A$ and $\Pi \leq P_A(\Lambda - f) \leq 0$ for $f \geq \Lambda$; and $\Lambda \leq \Lambda/((1-p_1)+p_1/F)$ since $F \geq 1$. Otherwise $\hat{c}(a) \geq 1$. If no targeted class uses $[k]$, then $\kappa \geq 1$. If one does, call it t_0 (injectivity allows at most one); then $\kappa \geq 1 - p_{t_0}(1-k/F) \geq 1 - p_1(1-1/F) = (1-p_1) + p_1/F$, using $k \geq 1$, $p_{t_0} \leq p_1$ and $F \geq 1$. In every case $\Pi \leq \Lambda - f\big((1-p_1)+p_1/F\big) \leq 0$.

(d) $2p_1 \geq p_1(1+1/F) + 2(1-p_1)/F \iff p_1(1+1/F) \geq 2/F \iff p_1 \geq 2/(F+1)$. □

Scope of A2. The plan class above, with an injective map and a dedicated abort set, is what A2 and A3 jointly imply: by A3 a fixed reveal set runs one pattern in every class, and by A2 running it outside its prepared class loses at least the fees $(k - |S|)f(1-1/F)$ saved relative to aborting. Theorem 3(b) is an explicit plan whose miss branch does nothing, so it does not depend on the size of misdirected losses. Parts (a) and (c) and the entries of Table 3 above Λ do use $\ell \geq f(1-1/F)$. In practice the intrinsic loss of a misdirected execution, such as swap fees or the gas of a reverting call, is set by market conditions and need not scale with f; if misdirected executions are only assumed not to gain, the bound of Section 8 is $f \geq Fp_1\Lambda$.

Interpretation. With one preceding commitment uncertainty protects the victim: the attacker forfeits the full fee on every miss, and $f_1(p_1)$, which has the form of the supportable-fee expression in the LUCID specification [32], is roughly $p_1\Lambda$ for small p_1. With two commitments a miss costs $2f/F$

N	$m=1$	$m=2$	$m=3$	$m=4$	$m=5$	$m=6$
4	.332/.332	.977/.948	1.303/.985	1.303/1.290	1.303/1.290	1.303/1.290
16	.067/.067	.895/.805	.895/.848	.895/.863	.895/.871	.895/.876
64	.016/.016	.670/.502	.670/.574	.670/.603	.670/.618	.670/.628
256	.004/.004	.334/.200	.402/.301	.419/.354	.431/.386	.440/.408

Table 3: Deterring fee f_m^*/Λ for a uniform prior over N classes, $F = 128$, in the general class / selector class.

and the full fee is paid only on a hit, so the deterring fee rises to at least $\Lambda/2$ once $p_1 \geq 2/(F+1)$, that is, unless the plaintext has more than about $\log_2 F - 1$ bits of min-entropy against the attacker (six bits for $F = 128$); at $p_1 = 1/64$ it is 32 times the single-commitment value (Table 2). By part (c) no number of commitments forces a fee above $f_1(p_1)/p_1$. A victim can set the cap to one, at the cost of competing for the top of the block, or post a fee close to the extractable value, which is held for a slot. The argument uses only that revealing costs a $1/F$ fraction of forfeiting, so it applies to any per-commitment pricing with cheap reveals.

Numerical evaluation. For fixed (k, a) and fixed f, the profit is maximized by assigning classes in decreasing order of p_t to reveal sets in increasing order of cost, keeping a targeted class only if its marginal gain over aborting, $\Lambda + c(a) - c(\sigma(t))$, is positive (rearrangement inequality; this is the rewriting $\Pi = -c(a) + \sum_{t \in A} p_t(\Lambda + c(a) - c(\sigma(t)))$, and it keeps classes whose reveal set costs more than Λ when the abort set costs more still); profit is decreasing in f, so f_m^* is found by bisection. We validated this procedure against exhaustive enumeration on 300 random instances in both classes, and confirmed the bounds of Theorem 3 on 3000 random priors with $F \in \{2, 8, 32, 128, 1024\}$ and $m \leq 6$; both bounds are attained on some instances. Table 3 reports f_m^* for uniform priors. The values change little beyond $m = 2$.

7 Option migration

Threshold release removes the sender's option by Theorem 1(a). We ask who holds the option afterwards.

Committee model. A committee $[n]$ holds shares of a decryption key with threshold $1 \leq t \leq n$. The batch executes iff at least t shares are published by t_2. A coalition $W \subseteq [n]$ acts strategically and the other $n - |W|$ members publish their shares before some time $t^* < t_2$. The coalition publishes $r \in \{0, \ldots, |W|\}$ of its own shares, so $X = \mathbf{1}\{(n - |W|) + r \geq t\}$. Its gain from non-execution is D_W, and an attributable penalty replaces D_W by $D_W - \beta$, where $\beta \geq 0$ is the net penalty the coalition incurs when it switches from $X = 1$ to $X = 0$; with per-share penalties, β is the cheapest such switch, which depends on how many members must withhold. We assume:

D1 (last mover) the coalition decides after the honest shares are visible to it;

D2 (public reconstruction) any t shares reconstruct the key without further interaction.

D2 is a standard requirement for mempool threshold schemes [41, 29]. D1 is a timing assumption: honest members publish promptly, while a coalition may wait until the deadline.

Theorem 4 (Option migration)**.** *(a) The coalition can force both $X = 0$ and $X = 1$ if and only if $|W| \geq n - t + 1$; if $|W| \leq n - t$, then $X = 1$ whatever it does.*

(b) *Under D1 and D2, at t^* the coalition holds $n \geq t$ shares and can reconstruct the key privately, so the plaintexts belong to its information at t^*. Every blocking coalition therefore has binary control on Ω at t^* with mechanism information, and by Theorem 1(b)*

$$V_W^{\mathrm{endo}} = \mathbb{E}\big[(D^{\mathcal{F}}_{W,t^*} - \beta)^+\big] - \mathbb{E}\big[(D^{\mathcal{G}}_{W,t^*} - \beta)^+\big] \geq 0,$$

while the sender's option is zero whenever the honest members alone reach the threshold.

(c) *A coalition with $|W| \geq t$ can decrypt before the ordering is fixed, and one with $|W| \geq n-t+1$ holds the option of (b). The smallest coalition able to do either has size $\min(t, n-t+1) \leq (n+1)/2$, with equality if and only if $2t = n+1$; for even n it is at most $n/2$. As a size it is an integer, so it is at most $\lceil n/2 \rceil$.*

(d) *If $\beta \geq D^{\mathcal{F}}_{W,t^*}$ almost surely, then $V_W^{\mathrm{endo}} = 0$. In general $V_W^{\mathrm{endo}} > 0$ if and only if $\Pr[D^{\mathcal{F}}_{W,t^*} > \beta \geq D^{\mathcal{G}}_{W,t^*}] + \Pr[D^{\mathcal{F}}_{W,t^*} < \beta \leq D^{\mathcal{G}}_{W,t^*}] > 0$.*

Proof. (a) $X = 0$ is reachable iff $n - |W| < t$; $X = 1$ is reachable iff $n \geq t$.

(b) By D1 the coalition holds $(n-|W|)+|W| = n \geq t$ shares at t^*; by D2 it reconstructs without publishing. Its choice of r is made at t^* with this information and only X affects payoffs, which is binary control on Ω with D replaced by $D_W - \beta$. The statement about the sender is Theorem 1(a).

(c) $\min(t, n-t+1) \leq \frac{1}{2}(t+n-t+1)$ with equality iff the two arguments agree; for even n, $n+1$ is odd so equality is impossible.

(d) If $D^{\mathcal{F}}_W \leq \beta$ then $D^{\mathcal{G}}_W \leq \beta$ and both terms of (b) vanish; the criterion is Lemma 2 applied to $D_W - \beta$. □

Corollary 3 (Hybrid release without independence)**.** *In the hybrid setting with an arbitrary outage event $R \in \mathcal{G}_{t^*}$, $\Pr[R] = 1-q$, let $Z = \mathbb{E}\big[(D^{\mathcal{F}}_{t^*})^+ - (D^{\mathcal{G}}_{t^*})^+ \mid \mathcal{G}_{t^*}\big]$ and $V_{\mathrm{sr}}^{\mathrm{endo}} = \mathbb{E}[Z]$. Then:*

1. *$Z \geq 0$, $V^{\mathrm{endo}} = \mathbb{E}[\mathbf{1}_R Z] = (1-q)V_{\mathrm{sr}}^{\mathrm{endo}} + \mathrm{Cov}(\mathbf{1}_R, Z)$, $|\mathrm{Cov}(\mathbf{1}_R, Z)| \leq \sqrt{q(1-q)}\,\mathrm{sd}(Z)$ and $0 \leq V^{\mathrm{endo}} \leq V_{\mathrm{sr}}^{\mathrm{endo}}$;*
2. *if $\mathcal{G}_{t^*}$ is atomless, the largest value of V^{endo} over outage events $R \in \mathcal{G}_{t^*}$ with $\Pr[R] = 1-q$ is the expectation of Z over its upper $(1-q)$-quantile;*
3. *if the sender can cause an outage at cost c, deciding with $\mathcal{G}_{t^*}$-information, its option is $\mathbb{E}[(Z-c)^+]$.*

Proof. By Theorem 1(b), $V^{\mathrm{endo}} = \mathbb{E}[\mathbf{1}_R((D^{\mathcal{F}})^+ - (D^{\mathcal{G}})^+)] = \mathbb{E}[\mathbf{1}_R Z]$ by the tower property, since $R \in \mathcal{G}_{t^*}$; $Z \geq 0$ by Lemma 2. The covariance identity is algebraic and the bound is the Cauchy–Schwarz inequality. Item 2 is the bathtub principle. For item 3 the sender chooses a $\mathcal{G}_{t^*}$-measurable trigger Y and earns $\mathbb{E}[Y(Z-c)]$; apply Lemma 1. □

Discussion. The option is conserved rather than destroyed: threshold release moves it to every blocking coalition, which sees the plaintexts before deciding. This is the last-mover unfairness of threshold decryption [15], which complete fairness cannot avoid without an honest majority [10] and which penalties for aborting compensate [8]; our contribution is to value it in the mempool setting and to show that no threshold keeps both harms beyond $\lceil n/2 \rceil$ members. For EIP-8105 with a single provider ($n = t = 1$) the provider is itself a blocking coalition [24]. Treating withholding as an availability attack that timeliness votes can penalize [25, 46] requires a penalty above the largest single gain, not an average, and such penalties also fall on honest members who are late. In ticket-first designs with signed refusals [29], a blocking coalition holds the option with $\beta = 0$ unless

refusals are penalized. Coalition-deterrence results price coalitions of size at least t that leak early [31]; blocking coalitions need a separate stake. For hybrid release, unpredictability of outages at submission time [30, 41] is not sufficient; what matters is their correlation with the option's value at decision time. Option value is concentrated in practice [27]: for a lognormal Z with $\sigma = 2$, outages affecting 1% of decisions can capture up to 37% of $V_{\mathrm{sr}}^{\mathrm{endo}}$ (64% at 5%), against 1% (5%) under independence.

8 Blind simultaneous release

The preceding sections locate the option in a single place: a party decides after the mechanism has released information. Corollary 1 suggests the remedy, which is to have every release decision taken before anything is released. Theorem 2 shows that timed cryptography cannot keep a ciphertext private across a mempool's privacy window. Here it is asked to do much less: privacy until the ordering is fixed is provided by the underlying scheme (threshold encryption, or encryption to a key publisher), and the time-lock only has to make release decisions simultaneous across a window of one or two seconds. The technique, commit with a forced-opening fallback, is the one used for fair exchange [9], resource-fair computation [14], fair secret reconstruction [16] and randomness beacons [7]. We adapt it to key and share release and analyse it with the tools of the previous sections.

Releasers. A *releaser* is a party whose action after the ordering is fixed determines whether, and how, committed ciphertexts are opened: a key publisher, whose release value v_i is the bundle of keys for its users' transactions in the slot; a committee member, whose release value is its partial decryption for the batch; or, in a hybrid design, a sender holding a fallback key. Release values are determined by data fixed at t_1. Each admits a public validity check: a key is valid if it opens its ciphertext under a key-committing authenticated encryption scheme, and a share is valid if it verifies against the member's public verification key.

Timed pads. Let H be a hash function, G a key derivation function and L a length. A releaser prepares, ahead of time, pairs (sd, y) with sd a fresh random seed and $y = H^L(\mathsf{sd})$. To lock a string x it outputs (sd, z) with $z = x \oplus G(y)$; anyone opens (sd, z) as $z \oplus G(H^L(\mathsf{sd}))$. Locking costs no online computation, opening is deterministic, and opening requires L sequential evaluations of H. Iterated hashing is sequential in the random oracle model, and proofs of sequential work based on it remain sound against quantum adversaries [18], so the pads are post-quantum. Time-lock puzzles from lattices with logarithmic encryption cost after preprocessing [6] are an alternative that avoids the precomputation, at the price of newer assumptions.

Definition 5 (Solver parity)**.** Let ν_h be the sequential speed, in evaluations of H per second, of the fastest honest solver on which the protocol relies, and let ν_a be the sequential speed of any adversary. The protocol satisfies γ*-solver parity* if $\nu_a \le \gamma\, \nu_h$.

We state solver parity as an explicit assumption. It holds with γ close to one if honest solvers, which only need to exist and be live, use hardware of the same class as the strongest adversary; a protocol can fund such solvers as a public good. We are not aware of a published measurement of the sequential latency of iterated SHA-256 on latency-optimized hardware relative to general-purpose processors. As a baseline, a single SHA-256 compression with hardware extensions takes 70 ns (146 cycles) per step on a 2.1 GHz Xeon in our measurement.

Protocol. Let ω be the escrow window, Θ the opening window, S a slashable deposit per releaser, and L the pad length. Times are as in Section 3: the ordering and the set of included ciphertexts are fixed at t_1.

1. *Escrow.* During $[t_1, d]$ with $d = t_1 + \omega$, each releaser broadcasts an escrow $e_i = (h_i, \mathsf{sd}_i, z_i)$, where $h_i = \mathsf{Com}(v_i; r_i)$ is a hiding and binding commitment and (sd_i, z_i) locks $v_i \| r_i$. A timeliness committee attests which escrows it received by d, and consensus fixes the set E of timely escrows, as availability is fixed in existing designs [29]. A releaser without an escrow in E has chosen not to release. Each escrow is signed by its releaser and bound to the slot. If two distinct escrows of one releaser reach the committee by d, its value is $\perp$ and it is excluded from E, a choice fixed at d; a second signed escrow after d does not change the outcome and is slashable by at least S. The seed sd_i is fresh and uniformly random, and the hash is domain-separated by the predictable slot index and the releaser, which prevents sharing between releasers and replay across slots; an outside party cannot open a pad early because it does not know the seed until the escrow is broadcast, while the releaser may precompute its own pad at any time.

2. *Fast reveal.* After d, each releaser publishes (v_i, r_i). A reveal is accepted if it opens h_i and v_i is valid.

3. *Forced opening.* For every escrow in E whose value is needed and was not accepted by fast reveal, any party computes the pad. If the result opens h_i, it is used as if revealed (or as $\perp$ if invalid). If it does not open h_i, the escrow is inconsistent: anyone can verify this by recomputation, the releaser forfeits S, and its value is $\perp$. In the committee setting, an escrow is needed only while fewer than t valid shares are available.

4. *Execution.* The batch executes at the top of the first payload whose deadline t_2 is at least $d + \Theta$, using the values available by then; values not available are $\perp$ and the corresponding transactions are not executed. No transaction executes between d and the batch except in payloads sealed before d.

The pad length satisfies $L/\nu_h \le \Theta$, so honest solvers open any timely escrow by t_2, and $L/\nu_a \ge \omega$, so no adversary opens any escrow before d. By Theorem 2(a), applied with the earliest escrow broadcast at t_1 and the latest at d, such an L exists if and only if $\gamma \le \Theta/\omega$.

After d, fast reveals make release values public, so any decision a releaser takes after d is taken with mechanism information. We write $t^* \in (d, t_2]$ for the time of such a decision.

Theorem 5 (Blind simultaneous release)**.** *Assume γ-solver parity with $\gamma \le \Theta/\omega$ and L as above, a hiding and binding commitment, and that before d no party publishes information about another party's release value. Then:*

(a) Each releaser's decision to release is fixed by its escrow, taken by time d when the mechanism has released no information, so its endogenous option from this decision is zero. After d, a releaser can change whether its value is used only by withholding an inconsistent escrow, at cost S; the resulting option is $\mathbb{E}[(D^{\mathcal{F}}_{t^*} - S)^+] - \mathbb{E}[(D^{\mathcal{G}}_{t^*} - S)^+]$, which is zero if $S \ge \operatorname{ess\,sup} D^{\mathcal{F}}_{t^*}$.

(b) Under the pricing of Section 6, let g be the largest gain, in any class, from changing an attacker's executed set after d, and let $S \ge g$. Then no such change is profitable, and every attack plan executes the same reveal set, with the same pattern, in every class. Under A2, every plan is unprofitable whenever $f \ge f_1(p_1)$, for every cap m. If misdirected executions are only assumed not to gain, $f \ge F p_1 \Lambda$ suffices.

(c) In the committee setting, before d a coalition with $|W| < t$ holds fewer than t shares and no honest share, so its escrow decisions are blind and it cannot decrypt. After d, a coalition can prevent execution only by withholding at least one inconsistent escrow, at cost at least S, and only if $|W| \geq n - t + 1$. If $S \geq \operatorname{ess\,sup} D^{\mathcal{F}}_{W,t^*}$, informed blocking is never profitable: every content-dependent harm, early decryption or informed blocking, requires $|W| \geq t$, and a coalition with $n - t + 1 \leq |W| < t$ can block only blind. With $t = \lceil 2n/3 \rceil$ such harm then requires $\lceil 2n/3 \rceil$ members, against at most $\lceil n/2 \rceil$ for any threshold without blind release (Theorem 4(c)); the first is never smaller and is strictly larger for $n = 2$ and every $n \geq 4$ (equal for $n \in \{1, 3\}$).

(d) In a hybrid design whose fallback keys are escrowed in the same window, the sender's decision at d carries no endogenous option whatever the correlation between committee outages and the option's value. If $S \geq \operatorname{ess\,sup} D^{\mathcal{F}}_{t^*}$, the sender's endogenous option is zero and only the exogenous option remains; otherwise it is at most the option of (a).

If solver parity fails with probability ε, all statements hold up to an additive $\varepsilon \|D\|_\infty$, by the argument of Theorem 1(a).

Proof. (a) Honest escrows are broadcast no earlier than t_1, and no adversary opens one before $t_1 + L/\nu_a \geq d$; commitments are hiding. Hence, up to time d, the mechanism information $\mathcal{H}$ contains nothing about release values, and a releaser's escrow is a function of $\mathcal{F}_d = \mathcal{G}_d$. Corollary 1 gives zero endogenous option for this decision. After d, consider a releaser with escrow in E. If its escrow is consistent, binding of h_i and determinism of the pad give the same value whether the releaser reveals or not, and honest solvers obtain it by $d + L/\nu_h \leq t_2$; the releaser has no control. If the committed value is invalid, binding again fixes the outcome ($\perp$) at d. If its escrow is inconsistent, it chooses at t^* between revealing the committed value and withholding at cost S, which is binary control with penalty S; Theorem 4(d) gives the stated value and the deterrence condition.

(b) By (a), changing the executed set after d requires withholding at least one inconsistent escrow, which costs at least $S \geq g$, so no deviation after d is profitable. The executed reveal set R is therefore fixed by the escrows at d, before the victim's value can be opened. Since programs read nothing about the plaintexts of later transactions (A3), the pattern that R executes is also the same in every class, so it is prepared for at most one guess class t (Section 6). If $R = \emptyset$, the plan executes nothing and its profit is $-kf \leq 0$. Otherwise $c(R) \geq f/F$; write $-\ell$, with $\ell \geq 0$ by A2, for the gross payoff of the pattern in a class it was not prepared for. Then

$$\Pi \leq p_t \Lambda - (1 - p_t)\ell - f/F.$$

With $\ell = 0$ this is at most $p_1\Lambda - f/F$, which is non-positive if $f \geq F p_1 \Lambda$. Under A2, executing R in a class it was not prepared for is no better than aborting all k commitments, so $-\ell - c(R) \leq -kf$, that is, $\ell \geq kf - c(R) = |R|\, f(1 - 1/F) \geq f(1 - 1/F)$, and $\Pi \leq p_t\Lambda - f\big((1 - p_t) + p_t/F\big)$. This is increasing in p_t, hence at most $p_1\Lambda - f\big((1 - p_1) + p_1/F\big)$, which is non-positive iff $f \geq f_1(p_1)$.

(c) Before d, honest shares exist only inside escrows that cannot be opened before d, so a coalition with $|W| < t$ cannot decrypt, and its escrow decisions are functions of $\mathcal{G}_d$. A coalition with $|W| \geq t$ can decrypt with its own shares. After d, every consistent escrow in E yields its value by t_2, and the only post-deadline control is withholding inconsistent escrows (part (a)). Let j members of W have a release value that is already unavailable at d (no escrow in E, or an invalid committed value), a choice made at d, and let k inconsistent escrows be withheld; then $n - j - k$ values are available and execution fails iff $n - j - k < t$. Since only members of W contribute to

$j + k$, failure requires $|W| \geq n - t + 1$ (Theorem 4(a)); if it is caused after d, then $k \geq 1$ and the cost is at least S. Informed blocking is thus binary control at t^* with penalty at least S, and Theorem 4(d) with $\beta = S$ gives zero option under the stated condition. The bound S cannot be raised to $(n-t+1)S$: with $j = n-t$, which by itself does not block, a single withheld inconsistent escrow does. For $t = \lceil 2n/3 \rceil$, $\lceil 2n/3 \rceil \geq \lceil n/2 \rceil$ for every n, with equality only at $n \in \{1, 3\}$.

(d) The fallback key is a release value escrowed by d, when $\mathcal{F}_d = \mathcal{G}_d$, so the variable Z of Corollary 3 at that decision is identically zero and $V^{\text{endo}} = \mathbb{E}[\mathbf{1}_R Z] = 0$ for every outage event R. The decision at t^* is covered by (a). □

Parameters. An escrow window ω of 1 to 2 seconds is realistic, since small consensus messages reach a supermajority in 0.83 s at the median and 1.96 s at the 95th percentile [19]. Under the enshrined proposer-builder separation timeline [13], executing the batch in the next payload gives $\Theta = 12 - \omega$ seconds, so $\gamma_{\max} = \Theta/\omega$ is 5 for $\omega = 2$ s and 11 for $\omega = 1$ s with twelve-second slots, and 2 and 5 with six-second slots. At 70 ns per hash step the pad has about 1.4×10^8 steps, prepared ahead of time (Appendix A).

Implementation requirements. The proofs assume signed, slot-bound escrows with domain-separated random seeds; a way to settle claims of inconsistency without recomputing the pad, for which we assume an interactive bisection game that we do not analyse; integrity of the escrow set under the timeliness committee; funded parallel solvers; and staked releasers (Appendix A).

Costs and limits. In the common case every releaser reveals in the clear, no pad is computed, and the only added latency is ω; withholding never delays the chain or empties a block, since unopened values are dropped. Four limits remain. Blind blocking, by not escrowing, is still possible for $n - t + 1$ members or a publisher for its own users, but it is decided at d without plaintexts, so its option is the exogenous one of Section 9. Informed blocking after d is deterred only by the deposit (Theorem 5(c)), which is forfeited only on a verifiable inconsistency and never for lateness; it must cover the largest single gain and locks capital. Coalitions of t or more members can still decrypt early. And pads cost about one core per releaser, which suits publishers and committee members but not end users.

9 Calibration

Inputs. Ethereum's implied volatility index stood at about 51% annualized in mid-October 2026, between 47.5% and 53.1% over the preceding month [23]; we also consider 100% and 150%, typical of high-volatility days. The LUCID specification names its deadlines (commitments fixed at the start of a slot, keys due before a timeliness vote in the next slot) without numeric values [32], so we treat the reveal window Δ as a parameter between 4 and 16 seconds; 8 seconds matches the builder-option window studied in [27]. We scale returns with $\sqrt{\Delta}$; at short horizons scaling lies between square root and linear [27], so this is the conservative choice.

The exogenous option of a sender. A sender whose sealed transaction is a trade of notional N gains $D \sim \mathcal{N}(0, s^2)$ from withholding, with $s = N\sigma\sqrt{\Delta/\text{year}}$, and withholding costs the additional forfeit $\theta = f(1 - 1/F)$. By Lemma 3 with $\mu = -\theta$, the option is $V(\theta) = s\varphi(\theta/s) - \theta(1 - \Phi(\theta/s))$, exercised with probability $1 - \Phi(\theta/s)$; the fee capping that probability at α is $f_\alpha = sz_{1-\alpha}/(1 - 1/F)$.

σ	Δ (s)	s	$V(0)$	$f_{1\%}$	$f_{1\%}/F$	$f_{0.1\%}$
51%	4	1.82	0.72	4.26	0.033	5.66
51%	8	2.57	1.02	6.02	0.047	8.00
51%	16	3.63	1.45	8.51	0.067	11.31
100%	8	5.03	2.01	11.81	0.092	15.68
150%	8	7.55	3.01	17.71	0.138	23.52

Table 4: The exogenous option and the fee that caps its exercise probability, in basis points of notional, $F = 128$. The column $f_{1\%}/F$ is the amount burned when the sender reveals.

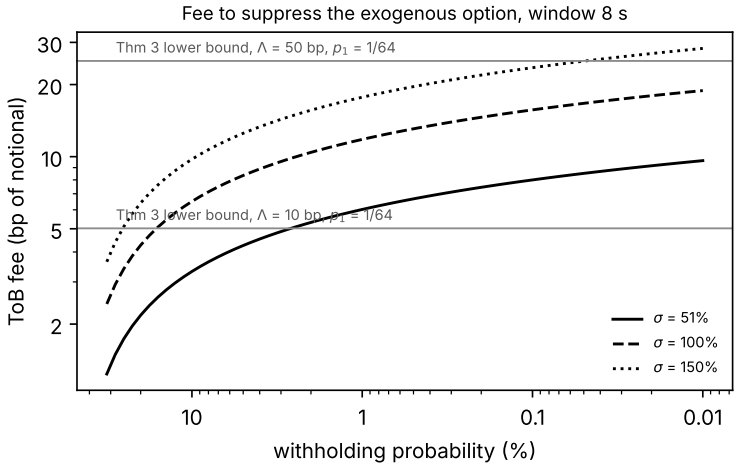


Figure 1: Fee capping the exercise probability of the exogenous option ($\Delta = 8$ s), against the lower bound $f_{\mathrm{inv}}(1/64) = 0.502\Lambda$ of Theorem 3 for attacker gains Λ of 10 and 50 basis points.

Results. At 51% volatility and an 8-second window, an unpenalized sender holds about one basis point of notional, \$102 on a \$1M trade (Table 4). A fee of about six basis points caps exercise at 1%, of which 0.05 basis points (\$4.70) is burned when the sender reveals. With a slippage tolerance of 0.5% an attacker extracts up to $\Lambda = 50$ basis points, and for caps $m \geq 2$ Theorem 3 requires a fee of at least 25 basis points when $p_1 = 1/64$, above the 1% exogenous fee even at 150% volatility (Figure 1); without blind release, probabilistic frontrunning is the binding constraint. Under blind release and A2 the frontrunning fee falls to $f_1(1/64) \cdot 50 \approx 0.8$ basis points and the exogenous option becomes binding. The exogenous fee scales with $\sigma\sqrt{\Delta}$ while the frontrunning fee depends only on Λ and p_1, so a volatility-indexed fee as proposed for the builder option [27] addresses only the first. Gaussian quantiles understate tails: a Student-t distribution with the same variance and 5 degrees of freedom (illustrative, not fitted) raises the 8-second fees at 51% volatility from 6.0 and 8.0 to about 6.8 and 11.8 basis points for 1% and 0.1% (15.3 with 3 degrees of freedom), still below 25.

10 Related work

Encrypted mempool constructions. Threshold-encrypted mempools originate with Shutter and Ferveo [39, 26]. Batched threshold encryption [21, 22], silent setup [40], their combination [5], and epochless and collision-free variants [43] address communication cost, setup and censorship;

Passerat-Palmbach surveys this line and the Ethereum deployment debate [41]. Garg et al. cast many of these schemes as witness encryption for linearly verifiable relations [45]. Our results are agnostic to the scheme: they concern who can cause opening and when.

Ethereum proposals. LUCID [32] and EIP-8105 [24] are the most developed in-protocol proposals; LUCID was not selected as a headliner for the Hegotá upgrade [3], and its specification remains under development [33]. Herold's critique identifies sender optionality as the main weakness of commit-reveal designs and argues that timed cryptography, trusted hardware and current threshold schemes do not resolve it [28]; his ticket-first threshold identity-based design avoids batching [29]. Wagner and Ma propose hybrid release [30]. Allowlists of key publishers are under discussion [46, 25]. Our contribution relative to these is quantitative and formal: a measure of the option, exact conditions for its absence, the fee-inversion attack, and the migration of the option to blocking coalitions.

Economics of mempool privacy. Rondelet and Kilbourn analyse the market effects of mempool privacy [37]. Garimidi, Bonneau and Heimbach discuss speculative attacks via non-decryption and argue that witness encryption subsumes committee and delay approaches [2]. Lin et al. characterize coalition deterrence against early leakage in threshold mempools [31]; withholding, which we study, is the complementary harm. Mazorra et al. introduce the free option of enshrined proposer-builder separation and propose penalties and window shortening [27]; our separation theorem and calibration apply their option-pricing view to the release of encrypted transactions.

Fairness and forced opening. Fair exchange and fair computation have long used timed commitments to stop a party from learning an output and then aborting [9, 14]; complete fairness is impossible without an honest majority [10], and penalties have been used to compensate honest parties [8]. Knapp and Quaglia use homomorphic time-lock puzzles for fair secret reconstruction [16], Kavousi et al. introduce timed secret sharing [20], and Bicorn combines commit-reveal with forced opening by a single time-lock puzzle to remove last-revealer bias from randomness beacons [7]. For threshold decryption, Hong et al. identify the last-mover unfairness and resolve it with semi-trusted parties [15]. Time-lock and computational decryption have been proposed for mempools as the primary privacy mechanism [17, 11]; Theorem 2 explains the tension this creates. Blind simultaneous release instead confines the time-lock to the decision window.

Timed cryptography and randomness. Time-lock puzzles [38], timed commitments [9, 42] and verifiable delay functions [44] provide forced opening without trusted parties; our timing bound applies to all of them. The last-revealer problem of commit-reveal randomness beacons, and its optimal exploitation by withholding, have been analysed with Markov decision processes [36, 35]; option migration is its counterpart for key shares.

11 Open problems

1. *Hardware gap.* Measure the sequential latency of iterated hashing on latency-optimized hardware relative to commodity processors, which determines the solver-parity factor γ that blind simultaneous release needs to stay below 5 to 11.

2. *Verifiable escrows.* A succinct proof that a hash-chain pad opens a given commitment would remove the dispute game and possibly the deposit; its proving cost is unknown.

3. *Batched post-quantum forced opening* of many threshold shares, to reduce solver work, for example from lattice-based batch time-lock puzzles [12].

4. *Pricing the residual.* The option under heavy-tailed returns, the optimal deposit when the essential supremum is infinite, and empirical estimation of the exogenous option.

5. *Equilibrium analysis.* We fix the strategies of other parties and bound coalitions in the worst case; a joint equilibrium of senders, publishers, committees and builders is open, and free-riding or coordination failure among non-colluding releasers would discount the withholding gains we bound.

6. *End-user escrows.* Whether senders without a staked publisher can take part in blind release, which requires removing both the capital barrier of the deposit and the pad-precomputation overhead, for instance with on-demand post-quantum time-locks without preprocessing.

7. *Formalization.* Extending the Lean development to general probability spaces, the L^p bound and the protocol layer.

12 Conclusion

The free option in encrypted mempools cannot be removed, only moved. Sound timely forced opening removes it from the sender, and the question is who else can decide after seeing information. Timed release answers this without trusted parties only if ciphertexts never wait in a public mempool and hardware advantages stay small; fee-based pricing fails against attackers with two commitments; threshold release and allowlists hand the option to blocking coalitions. Blind simultaneous release confines the option to its exogenous part by escrowing every release decision under a short time-lock, at the cost of an explicit assumption on honest solver speed (a margin of 5, or 11 with a median-latency window) and a stake sized to the largest single gain from informed blocking.

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A Parameters and implementation of blind simultaneous release

Parameters. The escrow window must let honest escrows reach the timeliness committee. Small consensus messages reach a supermajority of their subnet in under 830 ms at the median and in 1.96 s at the 95th percentile [19], so $\omega = 2$ s covers the 95th percentile and is the conservative choice, while $\omega = 1$ s covers the median only and is aggressive. Under the enshrined proposer-builder separation timeline, the execution payload is due six seconds into a twelve-second slot [13]. Opening the escrow window as soon as a payload is sealed and executing the batch in the next payload satisfies the execution rule with Θ equal to the slot length minus ω. With twelve-second slots this gives $\Theta = 10$ s and $\gamma_{\max} = \Theta/\omega = 5$ for $\omega = 2$ s, and $\Theta = 11$ s and $\gamma_{\max} = 11$ for $\omega = 1$ s; with six-second slots, $\Theta = 4$ s and $\gamma_{\max} = 2$ for $\omega = 2$ s, and $\Theta = 5$ s and $\gamma_{\max} = 5$ for $\omega = 1$ s. With commodity honest solvers at 70 ns per step, the pad length $L = \Theta\nu_h$ is about 1.4×10^8 for $\Theta = 10$ s and 1.6×10^8 for $\Theta = 11$ s; preparing one pad takes Θ seconds of one core, and one pad per releaser per slot suffices because a pad can lock the releaser's entire release value. Pads are prepared ahead of time, off the critical path.

Implementation requirements. Four points are needed for the proofs to apply and are not analysed further here. First, *verifying inconsistency.* A valid opening (v, r) of h_i is a short witness of consistency, so any solver that obtains one publishes it. The claim that an escrow is inconsistent has no short witness, because raw iterated hashing carries no succinct proof of sequential evaluation, and recomputing $L \approx 1.4 \times 10^8$ steps is beyond a validator's budget. We assume the claim is settled by an interactive bisection over checkpoints of the hash chain, as in optimistic verification of sequential computation, which reduces a dispute to one hash evaluation after $O(\log L)$ rounds (about 28 for $L \approx 1.4 \times 10^8$); a succinct proof of sequential work for the pad would remove the interaction. The dispute cannot finish within the opening window, so execution and slashing are separate: at t_2 an escrow in E with neither a fast reveal nor a valid opening is treated as $\perp$, and slashing is settled afterwards. The claimant posts the checkpoints and a bond; a claim stands unless challenged within a window; a valid opening of h_i posted before t_2 precludes the claim; a party that fails to answer a step in time loses by default. The deposit stays locked until the windows close,

so a releaser's collateral must cover its disputes in flight. Theorem 5 assumes that inconsistent escrows are identified and slashed. Second, *the escrow set.* Blindness at d rests on the timeliness committee not admitting an escrow after d: a supermajority colluding with a releaser could admit a late escrow chosen after seeing fast reveals, so the integrity of E is assumed as for availability in existing designs; excluding an escrow without seeing it is a blind choice, counted in j in the proof of (c). Third, *solver resources.* Honest solvers start every needed pad at d and stop when a reveal arrives, so reveal latency does not shorten Θ. Opening K escrows within Θ takes K cores in parallel at full clock speed, and a releaser can impose this cost by withholding consistent escrows at no protocol cost. In the committee setting at most t valid openings are needed; elsewhere the protocol must cap the number of distinct escrows per slot to bound K, and fund solvers, as the parity assumption already requires. Fourth, *deposits.* The deposit S is posted by releasers, that is, by key publishers and committee members; a single deposit covers the whole bundle of keys a publisher releases, and key-committing validity is checked per transaction once the bundle opens h_i, so one malformed ciphertext drops only its own transaction and does not slash the publisher. An attacker who splits its commitments across several releasers gains nothing in Theorem 5(b), since changing the executed set after d still requires withholding at least one inconsistent escrow, each costing at least S. A sender acting as its own publisher must post $S \geq g$ itself, which for the selector plan of Theorem 3(b) is about $\Lambda - f(1 - 1/F)$; senders without such a stake rely on a staked publisher who does not collude with them. In every case network delay is absorbed into ω: an escrow that misses d is simply not in E.

Costs and limits. In the common case every releaser reveals in the clear, no pad is computed, and the only added latency is ω. Forced opening runs only for withholders, and in the committee setting only until t valid shares are available; values not opened by t_2 are dropped, so withholding never delays the chain or empties a block. Four limits remain. First, blind blocking: a coalition of $n - t + 1$ members, or a key publisher for its own users, can still prevent execution by not escrowing; it decides at d without plaintexts, so its option is the exogenous one of Section 9, and it is attributable. Second, informed blocking after d is deterred only by the deposit (Theorem 5(c)). The deposit is forfeited only on a publicly verifiable inconsistency, never for lateness, so it can be set at the level Theorem 4(d) requires without taxing slow honest members; it must cover the largest single gain from informed blocking, and it locks capital. For unbounded payoffs the essential supremum is infinite and no finite S removes the option exactly; the residual is the tail expectation of Theorem 5(a), which a high-quantile S makes small. Third, coalitions of t or more members can still decrypt early; no threshold design avoids this. Fourth, a pad costs about one core per releaser, which suits publishers and committee members but not end users. Six-second slots halve Θ and leave a margin of 2 to 5, which makes parity with γ near one a requirement.

B Machine-checked proofs

We formalized the finite-probability versions of the results in Lean 4 (version 4.34.1) with Mathlib (v4.34.1). The development models a finite sample space with full support, information structures as labellings of outcomes, and conditional expectation as an explicit average over cells. It comprises 107 theorems; `#print axioms` reports only `propext`, `Classical.choice` and `Quot.sound` for each, and the sources contain no `sorry`. Table 5 lists the correspondence.

Code availability. The Lean sources, the audit output (`AXIOMS.txt`), a report on every deviation from the formalization briefs (`REPORT.md`), and the Python scripts behind the numerical tables, the

Paper	Formalized content
Lemmas 1, 2	bound and attainment; Jensen inequality with exact strictness criterion; non-strict parts also over general probability spaces
Theorem 1, Corollaries 1, 2	value identities, nonnegativity, positivity criterion, timing criterion; part (a) for bounded D with one or two good events; hybrid factor on an explicit product space
Theorem 2	part (a); both closed forms, strict monotonicity, the limit and the wait threshold in (b); monotonicity in (c); table entries
Theorem 3	(a) for one commitment; (b) as an explicit plan under every cap $m \geq 2$; (c) for all $k \geq 1$ over the general class; (d); table entries
Theorem 4, Corollary 3	committee combinatorics of (a) and (c) with the odd and even cases; reconstruction step of (b); penalized option value, strictness criterion and deterrence condition of (d); Corollary 3 items 1 and 3
Theorem 5	window condition $\gamma \leq \Theta/\omega$ as an instance of Theorem 2(a); margins and pad length; outcome of consistent and inconsistent escrows; zero option before d and the S-penalized option after d in (a); no profitable deviation in (b) and the fee bound for a fixed reveal set, in both forms; $Z = 0$ under equal information in (d); in (c), the counting behind the deposit condition and its tightness, and the comparison of $\lceil 2n/3 \rceil$ with every threshold without blind release, strict iff $n = 2$ or $n \geq 4$

Table 5: Correspondence between the paper's results and the Lean development.

fee-inversion search and the figure are available at `https://github.com/aryaethn/optionality`.

For Theorem 5, the protocol-level steps are not formalized: hiding of the escrows and of the commitment, determinism and sequentiality of the pads, solver parity (only the ratio ν_a/ν_h appears), consensus on the escrow set, the execution rule, and the step from the protocol to an executed set that is fixed at d with the same pattern in every class. The formalization also does not cover: the L^p form of Theorem 1(a); strictness criteria without full support; independence-based statements in their abstract form; the atomless statement of Corollary 3; the step from the timing assumptions D1 and D2 to binary control; Lemma 3; and the numerical evaluations.